

When Simple Wins: Leakage-Free Benchmarking of Classical Machine Learning vs. Graph Neural Networks in Low-Data Bioactivity Prediction^{*}

Andrey Vinajera-Zamora^a, Samuel Montejó-Sánchez^{a,*}, Raul Arias-Carrasco^a and Juan Castillo-Garit^a

^a*Instituto Universitario de Investigación y Desarrollo Tecnológico, Universidad Tecnológica Metropolitana, Ignacio Valdivieso 2409, Región Metropolitana, 97279, Santiago, Chile*

^b*Departamento de Informática y Computación, Universidad Tecnológica Metropolitana, José Pedro Alessandri 1242, Ñuñoa, Región Metropolitana, 7750000, Santiago, Chile*

ABSTRACT

Predictive modeling in low-data bioactivity regimes faces persistent methodological challenges, including high susceptibility to overfitting, silent data leakage, and unwarranted architectural complexity. Here, we present a leak-free benchmarking framework evaluating classical machine learning stacking ensembles against state-of-the-art Graph Neural Networks (ChemProp D-MPNN) for the inhibition of *Toxoplasma gondii* dihydrofolate reductase (TgDHFR, $N = 756$). Utilizing a rigorous 25-fold nested cross-validation protocol, we systematically compared four molecular representation spaces spanning 2D topological fingerprints, 2D physicochemical descriptors, and 3D geometric conformers. Classical stacking ensembles trained on lightweight Morgan fingerprints achieved superior predictive performance ($R^2 = 0.7417 \pm 0.0493$, MAE = 0.4841), statistically matching the performance of computationally intensive 2D/3D hybrid descriptors ($R^2 = 0.7348$, $p = 0.6471$) while entirely eliminating the conformational overhead of 3D modeling. Conversely, the D-MPNN underperformed under identical cross-validation conditions ($R^2 = 0.6818 \pm 0.0555$, MAE = 0.5329), demonstrating that data scarcity severely restricts end-to-end representation learning. Deploying the optimized classical model to screen FDA-approved therapeutics prioritized high-potency antifolates alongside high-efficiency scaffolds, including triamterene (ligand efficiency = 0.49) and pyrimethamine (ligand efficiency = 0.52). Subsequent orthogonal validation via AutoDock Vina molecular docking resolved screening discordances, successfully filtering out false positives such as bisacodyl while confirming high-affinity consensus leads, most notably Pralatrexate (docking energy: -9.43 kcal/mol). These findings demonstrate that in data-constrained QSAR regimes, the explicit inductive bias of fixed structural fingerprints acts as a natural regularizer that outperforms deep graph representations, providing a reproducible and computationally efficient paradigm for target-specific drug repurposing.

Keywords: Quantitative Structure-Activity Relationship, Low-Data Regimes, Nested Cross-Validation, Data Leakage, Graph Neural Networks, Molecular Docking

1. Introduction

1.1. Predictive Modeling in Low-Data Bioactivity Regimes

Predictive modeling in quantitative structure-activity relationships (QSAR) often adheres to the principle of parsimony, particularly where data is scarce Winkler (2026). In data-limited scenarios, simpler

^{*} smontejo@utem.cl

 avinajera@utem.cl (A. Vinajera-Zamora); smontejo@utem.cl (S. Montejó-Sánchez); raul.arias@utem.cl (R. Arias-Carrasco); j.castillo@utem.cl (J. Castillo-Garit)

 <https://orcid.org/0000-0002-4922-457X> (A. Vinajera-Zamora); <https://ieeexplore.ieee.org/author/37085682270> (S. Montejó-Sánchez); <https://orcid.org/0000-0003-1698-9282> (R. Arias-Carrasco); <https://orcid.org/0000-0003-0896-9484> (J. Castillo-Garit)

ORCID(s):

interpretable approaches like multivariate linear regression have traditionally prevailed, often demonstrating prediction reliability equivalent to complex artificial neural networks Shalit Peleg and Milo (2023); Xie, Wiriyarattanakul, Rungrotmongkol, Shi, Wiriyarattanakul and Maitarad (2023). These linear models maintain higher bias and lower variance, avoiding the extreme sensitivity to random fluctuations seen in complex algorithms, especially when supplemented by rigorous quantum chemical calculations to derive robust descriptors Xie, Wiriyarattanakul, Rungrotmongkol, Shi, Wiriyarattanakul and Maitarad (2023).

Modeling small datasets is inherently susceptible to overfitting, a bottleneck severely exacerbated when the high dimensionality of molecular descriptors vastly exceeds the sample size Hawkins (2004); Goodarzi et al. (2013). To combat this data sparsity, researchers frequently implement data augmentation (DAU) or synthetic sample generation to expand the training space Gerz and Jelali (2026); Meyer et al. (2025). However, these techniques introduce non-negligible computational complexity and resource demands, often requiring high-performance hardware and iterative optimization cycles Meyer et al. (2025); Calle et al. (2025). More critically, while DAU is designed to enhance model learning, its improper application often masks the true efficacy of the base algorithms by artificially elevating performance metrics through latent data dependencies and process-induced leakage Gerz and Jelali (2026); John et al. (2025).

Applying complex, non-linear architectures to these regimes introduces significant validation challenges. When augmentation is applied globally before dataset partitioning ("augment-then-split"), it violates core independent and identically distributed (i.i.d.) assumptions, allowing models to memorize augmentation artifacts rather than learn genuine physical rules Gerz and Jelali (2026); Veetil et al. (2024). This danger extends directly into model selection; tuning hyperparameters is exceptionally prone to overfitting when sample sizes are small and the parameter space is large Cawley and Talbot (2010). Consequently, evaluating machine learning architectures in isolation is meaningless without rigorously assessing the combined model selection procedure, making double (or nested) cross-validation mandatory to ensure true independence between model building and selection Baumann and Baumann (2014); Cawley and Talbot (2010).

Traditional evaluation metrics and standard k-fold cross-validation often misleadingly overestimate explorative prediction capabilities Király et al. (2022); Xiong et al. (2020). Data augmentation strategies can exacerbate underlying data structures, yielding overly optimistic accuracy metrics that mislead stakeholders regarding the model's true explorative power John et al. (2025). Furthermore, algorithms like Random Forest fundamentally lack extrapolation capability Xiong et al. (2020), and generative deep learning faces severe limitations with sparse, discrete chemical spaces like Simplified Molecular-Input Line-Entry System (SMILES) strings Liu et al. (2023). Therefore, to safeguard methodological integrity, any generative or augmentative process must be confined strictly to the training portion of the folds after a rigorous, independent partition has been established Gerz and Jelali (2026); John et al. (2025).

To safely harness the potential of overparameterized models Winkler (2026) without falling into the overfitting trap, rigorous validation protocols are mandatory. Techniques such as y-randomization (response shuffling) become essential to verify that the model captures true chemical relationships rather than random noise Rücker et al. (2007). Conversely, when descriptors are meticulously curated and validated, machine learning can effectively guide rational design, successfully discriminating potent bioactives—such as Amyloid- β aggregation inhibitors—from decoys Bansal et al. (2025).

Methodologies like transfer learning bypass data scarcity by adapting base networks previously optimized on large source domains Espley et al. (2023). By utilizing reduced learning rates during fine-tuning Bodun et al. (2025), and preferring efficient architectures like convolutional neural networks over heavier Transformer models Özçelik and Grisoni (2024), researchers can robustly navigate class imbalance and model selection bias in low-resource settings.

1.2. Domain-Engineered Features versus End-to-End Representation Learning

Traditional feature-based machine learning relies heavily on domain-engineered descriptors and molecular fingerprints Das et al. (2026). Methods like Morgan (ECFP4) encode structural traits, while topological, 3D, and physicochemical descriptors—such as radial distribution functions, Burden eigenvalues, and ab initio quantum chemical calculations—capture specific molecular interactions deterministically without requiring massive data Bansal et al. (2025); Iwuala et al. (2026); Xie et al. (2023). This approach embeds theoretical chemistry directly into the model, ensuring high interpretability and clarity Xie et al. (2023). However, accumulating these handcrafted features creates highly dimensional, redundant matrices Goodarzi et al.

(2013); Shalit Peleg and Milo (2023). In small datasets, this redundancy increases computational costs and severely worsens the feature selection bottleneck Goodarzi et al. (2013); Shalit Peleg and Milo (2023).

Alternatively, end-to-end representation learning bypasses manual feature selection by autonomously mapping discrete molecular topologies into continuous latent spaces Basit Iqbal et al. (2025); Azad (2026). Deep learning paradigms—including Autoencoders (VAEs), Generative Adversarial Networks (GANs), and Graph Neural Networks (GNNs)—extract task-specific traits directly from chemical structures to model complex, nonlinear interactions Azad (2026); Das et al. (2026); Liu et al. (2023). Similarly, Chemical Language Processing (CLP) applies natural language techniques to string representations like SMILES Özçelik and Grisoni (2024). However, learning representations in discrete chemical spaces is intrinsically fragile; altering even a single character in a SMILES string can irreparably invalidate the entire molecular structure Liu et al. (2023).

Despite these advantages, the utility of deep learning complexity in low-data regimes remains highly contested. These algorithms are inherently data-hungry, making representation learning highly ineffective when ground-truth labels and dataset volumes are severely restricted Konstantakos et al. (2025). To circumvent this limitation in the specific domain of *Toxoplasma gondii* dihydrofolate reductase (TgDHFR) inhibition, recent studies have resorted to highly complex pipelines. For instance, Karimova and Senanayake (2025) employed Gaussian noise-based data augmentation and deep neural network (DNN) ensembles to stabilize training on a restricted dataset, elevating their baseline R^2 from 0.75 to 0.85. This raises a fundamental research question: is such statistical complexity and synthetic data generation strictly necessary to achieve competitive predictive reliability, or can a parsimonious classical machine learning ensemble achieve equivalent baseline performance ($R^2 \approx 0.75$) “out-of-the-box” using standard domain-engineered features? Because small datasets cannot naturally sustain deep architectures without risking catastrophic overfitting Shalit Peleg and Milo (2023); Dar et al. (2023); Espley et al. (2023), simpler classical ensembles frequently achieve equivalent or superior predictive reliability in data-scarce scenarios Iwuala et al. (2026); Das et al. (2026); Xie et al. (2023).

Nevertheless, traditional skepticism toward nonlinear models in limited-data scenarios is changing Dalmau et al. (2025). When properly tuned, regularized, and hybridized with structured linear priors or heuristics to stabilize training, nonlinear algorithms can achieve competitive results comparable to robust classical methods Dalmau et al. (2025); Dar et al. (2023). Furthermore, transfer learning serves as a vital catalyst. By fine-tuning pretrained networks like ChemBERTa, researchers can achieve accurate representation learning predictions even with only dozens of target data points Oulladji et al. (2025); Espley et al. (2023).

1.3. Methodological Rigor and the Threat of Data Leakage in Molecular Modeling

Data leakage—the unintended introduction of test data information into the training process—is a statistically silent error that fundamentally undermines the validity of machine learning evaluations by inflating metrics without triggering code warnings Vrba et al. (2026); Gerz and Jelali (2026). In tabular molecular descriptor pipelines, this phenomenon typically manifests when global feature scaling (e.g., z-score standardization or MinMax scaling), descriptor imputation, or feature-filtering procedures (such as variance-threshold filtering or supervised feature selection) are executed prior to fold partitioning Correa et al. (2025); Lewis et al. (2023). Applying scaling globally calculates dataset-wide statistics, such as the mean and standard deviation, which inadvertently leaks information regarding the test set distribution into the training process. Consequently, this violates the fundamental independence assumption between partitions and yields overoptimistic, biased performance estimates Cawley and Talbot (2010). Similarly, executing feature selection or variance-based filtering over the entire dataset introduces a selection bias, as the chosen predictors retain a partial “memory” of the hold-out test data Varma and Simon (2006).

To eliminate these vulnerabilities, preprocessing, scaling, and feature selection must be treated as integral components of the model-fitting workflow Cawley and Talbot (2010). All transformation parameters must be strictly computed within each inner training fold and subsequently applied to the validation and test folds without parameter re-estimation Vrba et al. (2026). Rigorous partitioning protocols, such as strict three-way splits, must also be enforced to guarantee absolutely no overlap among training, validation, and test sets across subjects or discrete samples Dar et al. (2023).

Beyond preprocessing, data leakage frequently corrupts the model selection phase. Tuning hyperparameters globally across the entire dataset grants the algorithm illicit knowledge of the test distribution, yielding spuriously high performance Veetil et al. (2024). Even in traditional internal validation, an overlap between the data used for hyperparameter tuning and the test set creates an optimistic bias that renders the evaluation uninformative regarding real-world generalizability John et al. (2025). The validation dataset fundamentally loses its independence if its collective predictions continuously influence the search for an optimal model, triggering severe model selection bias. This lack of independence causes models—especially highly parameterized deep learning architectures deployed on small pretraining datasets Konstantakos et al. (2025)—to overfit by adapting to noise, producing deceptively overoptimistic internal metrics that severely underestimate true generalization errors Baumann and Baumann (2014).

To mitigate these structural biases, model selection must operate as an encapsulated component of the fitting process. Double or nested cross-validation is mandatory to generate strictly independent test objects in an outer loop that remain entirely isolated from model building and hyperparameter optimization Baumann and Baumann (2014). To further prevent fortuitous or biased splitting, researchers can employ deterministic methods like the Kennard-Stone algorithm to construct highly representative, unbiased external test sets Xie et al. (2023). When low-data regimes necessitate data augmentation, methodological purity remains paramount: classifier performance must be measured exclusively on real test data, keeping synthetic samples strictly isolated, while employing probabilistic Bayesian optimization rather than exhaustive grid searches to safely navigate the hyperparameter space Meyer et al. (2025). Preserving and transparently documenting this absolute methodological rigor is critical; insufficient reporting of data partitioning often confounds scientific literature, leading researchers to incorrectly attribute high performance to novel complex architectures when the success actually stems from silent data leakage Rojas and Madanian (2026).

1.4. Objectives and Principal Contributions

This study bridges the gap between stringent statistical validation and advanced representation learning to accelerate targeted drug discovery in low-data regimes.

- **Methodological Reproducibility and Leak-Free Validation:** Implementation of a fully transparent, 100% reproducible Python pipeline built upon a rigorous 5×5 repeated nested cross-validation protocol (25 outer evaluations). This ensures absolute statistical independence between hyperparameter optimization and model evaluation, effectively eliminating model selection bias and data leakage.
- **Architectural Benchmarking in Data-Limited Regimes:** A comprehensive empirical comparison evaluating the predictive reliability of traditional domain-engineered classical ensembles against state-of-the-art Graph Neural Networks (specifically, the ChemProp Directed Message Passing Neural Network, D-MPNN) using a standardized reference dataset.
- **Empirical Validation of Parsimony over Complexity:** Explicit demonstration that simple classical machine learning ensembles achieve statistical parity ($R^2 \approx 0.75$) with baseline deep learning models reported in recent literature. This establishes that competitive predictive reliability can be attained "out-of-the-box," effectively eliminating the necessity for synthetic data augmentation and computationally expensive 3D geometric preprocessing in low-data regimes.
- **Translational Drug Repurposing Pipeline:** The seamless integration of machine learning predictions with structure-based Molecular Docking. This hybrid pipeline systematically screens, identifies, and prioritizes FDA-approved compounds, providing a validated computational pathway for rapid therapeutic repurposing.

2. Materials and Methods

An overview of the multi-tier virtual screening pipeline implemented in this study is illustrated in Figure 1.

2.1. Bioactivity Benchmark Dataset and Curation

This study utilizes a previously published benchmark of *Toxoplasma gondii* dihydrofolate reductase (TgDHFR) enzyme inhibitors Karimova and Senanayake (2025). By adopting the exact baseline dataset

compiled by Karimova and Senanayake, we establish a strict methodological control. This framework isolates the predictive capacity of classical machine learning algorithms from computationally heavy statistical interventions. Deliberately omitting synthetic data augmentation during the primary benchmarking phase ensures that performance metrics reflect empirical ground-truth data rather than algorithmic artifacts.

Initial records merged drug-protein interaction parameters and SMILES structures from BindingDB with bioactivity data from ChEMBL Karimova and Senanayake (2025). Automated standardization protocols, including salt stripping and structural validation, ensured chemical integrity Bodun et al. (2025). To mitigate experimental noise and prevent overfitting Cherkasov et al. (2013); Iwuala et al. (2025), 117 compounds were systematically excluded from the initial 873 entries. This exclusion targeted organometallic complexes—which preclude 3D force-field generation—and molecules with undefined stereocenters that introduce severe conformational ambiguity.

For compounds exhibiting multiple IC_{50} measurements across varying experimental conditions, we applied the Interquartile Range (IQR) method. Duplicate assays with a bioactivity variance exceeding 1.5 log units were discarded as statistical outliers Karimova and Senanayake (2025). The remaining consistent entries were averaged to yield a single representative measurement per ligand.

This curation protocol yielded a final benchmark of $N = 756$ unique small molecules. The continuous target variable was transformed into pIC_{50} ($-\log_{10}(IC_{50})$) Karimova and Senanayake (2025). This logarithmic conversion standardizes response values and compresses the active range, optimizing the target distribution for robust regression modeling.

To explicitly quantify the performance impact of synthetic data generation reported in the original literature, we implemented an isolated data augmentation protocol exclusively for post-benchmark ablation studies. Following a strict 15% hold-out partition to prevent data leakage, training features underwent perturbation via additive Gaussian noise. Applying two distinct noise levels ($\sigma = 0.01$ and $\sigma = 0.001$) generated two synthetic variants per original molecule. This procedure effectively triplicated the training space while maintaining absolute isolation of the empirical test set.

2.2. Representation Spaces: Molecular Fingerprints and Conformational Descriptors

To systematically evaluate the structure-activity relationship, the open-source cheminformatics library RDKit Landrum et al. (2026) generated four distinct molecular representation spaces. The purely structural baseline utilized Morgan fingerprints (ECFP4), computed with a radius of 2 and folded into a fixed-length vector of 2048 bits. Extended-connectivity fingerprints (ECFPs) offer rapid calculation and capture an essentially infinite array of distinct molecular features, including stereochemical topology Rogers and Hahn (2010). Classical 2D fingerprint-based models frequently achieve predictive parity with state-of-the-art 3D structure-based models when relying exclusively on ligand information Gao et al. (2020).

To establish a quantitative physicochemical baseline, we extracted a comprehensive set of 217 conventional two-dimensional topological descriptors. Curated 2D molecular representations demonstrate substantial robustness; recent studies confirm they maintain scaffold-independent advantages over 3D alternatives across various property prediction tasks Tiwari et al. (2026). To evaluate the synergistic potential of combining structural and physicochemical data, a third representation space concatenated the 217 2D descriptors with the 2048-bit Morgan fingerprints.

To assess the predictive value of spatial geometry, a fourth hybrid representation incorporated 3D conformational descriptors. The Experimental-Torsion Knowledge Distance Geometry (ETKDGv3) approach generated the three-dimensional conformers Riniker and Landrum (2015). Because purely distance-based constraints often distort aromatic rings and sp^2 centers, ETKDGv3 integrates experimental torsion-angle preferences derived from small-molecule crystallographic data to optimize conformation generation Riniker and Landrum (2015). Following embedding, the MMFF94s force field minimized conformer energy, ensuring thermodynamic stability prior to 3D spatial descriptor extraction. Although recent empirical evaluations suggest that 3D conformer descriptors yield limited incremental predictive value over curated 2D representations in standard tabular benchmarks Tiwari et al. (2026), integrating spatial information remains critical for molecular image-based modeling Matsuzaka and Uesawa (2023) and geometry-augmented representation learning Zhang and Bai (2024).

2.3. Nested Cross-Validation Protocol and Variance Correction

Standard cross-validation partitions datasets into k disjoint blocks, systematically rotating them between training and testing phases Calle et al. (2025); Krstajic et al. (2014). While conventional nested cross-validation (NCV) literature frequently defaults to symmetrical 5×5 or 10×10 architectures Calle et al. (2025); Lewis et al. (2023); Wainer and Cawley (2021), methodological frameworks do not mandate identical fold counts for inner and outer loops Krstajic et al. (2014). To guarantee leak-free evaluation, balance computational overhead, and mitigate model selection bias in this low-data regime ($N = 756$), we implemented a strict 5×5 repeated nested cross-validation protocol (a 5-fold outer loop repeated 5 times, with a 5-fold inner loop). This architecture mathematically decouples model assessment from model selection via two distinct concentric loops.

Data partitioning strategy dictates the validity of small-scale QSAR modeling Király et al. (2022). Although Bemis-Murcko scaffold splitting is widely advocated to test out-of-domain extrapolation and prevent structural leakage Yang et al. (2019); Basit Iqbal et al. (2025); Iwuala et al. (2026), we deliberately selected a random nested cross-validation scheme. In a highly restricted dataset of 756 compounds, scaffold splitting frequently generates severely imbalanced folds. This creates artificial “activity cliffs” where entire bioactivity ranges vanish from the training pool, preventing the inner optimization loop from converging on generalizable hyperparameters. Because our primary objective is interpolation within the established applicability domain of known TgDHF_R inhibitors—rather than extrapolation to entirely novel chemotypes—random splitting ensures the bioactivity distribution remains statistically consistent across all folds. This consistency enables robust, stable hyperparameter tuning.

The outer loop of the NCV framework evaluates generalization. To achieve lower variance in performance estimation compared to extreme approaches like Leave-One-Out (LOO) Krstajic et al. (2014), we evaluated 25 outer splits. However, repeated cross-validation inherently introduces intra-repetition correlation, violating the strict independence assumption required by standard parametric tests. To rigorously account for this overlapping training data across folds, we applied the Nadeau and Bengio variance correction Nadeau and Bengio (1999) for all downstream statistical comparisons. In each iteration, one fold is strictly held out as the test set while the remaining pool passes into the inner loop. This absolute isolation ensures the test data remains hidden during feature and model selection, yielding a truly unbiased out-of-sample prediction estimate Varma and Simon (2006); Wainer and Cawley (2021).

The inner loop executes architecture selection and model optimization. Within each outer training pool, an embedded 5-fold inner loop evaluated all single, pairwise, and three-way stacking combinations (up to 3 base estimators). These combinations were drawn from six diverse regressor families: Random Forest (RF), Extra Trees (ET), Light Gradient Boosting Machine (LightGBM), eXtreme Gradient Boosting (XGBoost), Support Vector Regression (SVR with RBF kernel, $C = 10$), and k -Nearest Neighbors (kNN with cosine distance). For multi-model combinations, a `StackingRegressor` utilizing `RidgeCV` as the meta-estimator coordinated meta-learning. By confining model selection, descriptor scaling (`StandardScaler`), median imputation, and positive-importance feature selection entirely within the inner partitions, we prevent inadvertent information leakage. This strict encapsulation guarantees that reported metrics reflect genuine generalizability rather than selection artifacts Parvande et al. (2020); Lewis et al. (2023); Calle et al. (2025).

2.4. Deep Learning Baseline: Directed Message Passing Neural Networks

To benchmark the classical machine learning ensembles against advanced representation learning, we implemented a Directed Message Passing Neural Network (D-MPNN) using the Chemprop framework (version 2.0) Graff et al. (2025); Yang et al. (2019). To guarantee methodological parity, the D-MPNN underwent the exact 25-fold cross-validation protocol utilized for the classical baselines. Executing rigorous resampling on deep learning architectures incurs substantial computational overhead; however, this alignment ensures the resulting performance metrics remain directly comparable and free from single-split variance artifacts.

The architecture processes molecular SMILES strings by mapping them into graph representations via RDKit, designating atoms as vertices and bonds as edges Heid et al. (2023). Initial feature vectors capturing atomic identity and topological connectivity were constructed using `SimpleMoleculeMolGraphFeaturizer()`. The network applied `BondMessagePassing()` to extract embeddings directly from directed edges rather than vertices. This edge-centric design eliminates spurious feedback loops (totters), reducing noise in the

learned molecular representations Yang et al. (2019). A `MeanAggregation()` function pooled these atomic embeddings into a unified molecular vector.

Prior to training, the continuous pIC_{50} targets required internal normalization. The aggregated embeddings fed into a `RegressionFFN()` equipped with batch normalization to yield the final predictive output. Model weights were optimized via PyTorch Lightning over a maximum of 100 epochs. Unlike the classical ensembles, which underwent extensive hyperparameter optimization, the D-MPNN utilized default Chemprop hyperparameters (e.g., hidden size, depth, and learning rate). This methodological asymmetry stems from the prohibitive computational cost of nested cross-validation on graph neural networks using CPU infrastructure, coupled with the high risk of memorization when tuning overparameterized models on small datasets Cawley and Talbot (2010); Shalit Peleg and Milo (2023). Consequently, the D-MPNN represents an unoptimized, out-of-the-box deep learning baseline.

The internal loss function optimized during training was Root Mean Square Error (RMSE). Out-of-sample performance was quantified using the coefficient of determination (R^2) and Mean Absolute Error (MAE) averaged across all 25 folds. This rigorous statistical protocol yielded a final cross-validated test performance of $R^2 = 0.6818 \pm 0.0555$ and an MAE of 0.5329, establishing the deep learning baseline for subsequent comparative analysis.

2.5. Virtual Screening Protocol and Ligand Efficiency Profiling

An automated virtual screening pipeline, driven by the highest-performing classical ensemble (Morgan, nested CV $R^2 = 0.7417$), systematically prioritized viable lead candidates from an FDA-approved library. Rather than relying solely on raw predicted bioactivity, the evaluation framework integrated strict applicability domain enforcement with multi-objective ligand efficiency profiling. This approach mitigates the risk of prioritizing “molecularly obese” compounds that exhibit high theoretical potency but possess unfavorable physicochemical properties for downstream development Hopkins et al. (2004).

Initial structural pre-filtering restricted the FDA library to compounds matching the exact elemental composition of the training set (H, C, N, O, F, P, S, Cl, Br, I). This constraint prevents the model from extrapolating into unsupported chemical spaces. A strict physicochemical ceiling (molecular weight ≤ 1000 Da) controlled for collinearity, preventing molecular size from artificially inflating potency predictions.

To enforce the applicability domain, an `IsolationForest` algorithm (5% contamination threshold) trained on the original Morgan fingerprint matrix excluded structural outliers. This unsupervised anomaly detection ensured all advanced candidates resided strictly within the interpolated chemical space of the known bioactivity distribution. The ensemble’s predictive accuracy within this domain was internally validated by accurately recovering the pIC_{50} values of known reference inhibitors, such as Pyrimethamine and Trimethoprim. Ultimately, this hierarchical funnel successfully reduced the initial library to 17 prioritized compounds (10 top-ranked novel candidates and 7 established reference controls) for downstream orthogonal validation.

Candidate prioritization subsequently incorporated four distinct efficiency metrics to correct *in vitro* activity for physicochemical load. Ligand Efficiency (LE) quantified the binding energy per heavy atom count (HAC), requiring candidates to demonstrate optimal atomic utilization Hopkins et al. (2014):

$$LE = \frac{1.37 \times pIC_{50}}{HAC} \quad (1)$$

To account for heteroatom contributions and total mass, the Binding Efficiency Index (BEI) substituted HAC with molecular weight (MW) Abad-Zapatero and Metz (2005):

$$BEI = \frac{pIC_{50}}{MW} \times 1000 \quad (2)$$

Furthermore, to balance potency against lipophilicity and mitigate the risk of nonspecific binding, Lipophilic Ligand Efficiency (LLE) was computed as the difference between predicted potency and the calculated partition coefficient ($LLE = pIC_{50} - cLogP$). Finally, the Surface Efficiency Index (SEI) evaluated potency relative to the topological polar surface area (TPSA), providing a proxy for membrane permeability potential ($SEI = (pIC_{50}/TPSA) \times 100$). Because composite indices combining logarithmic potency values with linear physical parameters can introduce size-dependent mathematical artifacts Shultz (2013), these four metrics

(LE, BEI, LLE, SEI) operated strictly as secondary multi-objective ranking filters rather than primary objective functions.

2.6. Orthogonal Physics-Based Validation via Molecular Docking

Orthogonal structure-based evaluation via AutoDock Vina v1.2.0 Das et al. (2026); Trott and Olson (2010) complemented the ligand-based predictions. This physical validation translates abstract chemical representations into spatial coordinates, mitigating the risk of false positives induced by algorithmic scaffold memorization. By ensuring genuine structural complementarity, the docking protocol acts as a thermodynamic filter. Vina employs an Iterated Local Search optimizer and a hybrid scoring function, calibrated against PDBbind data, to guide flexible ligands into optimal binding conformations Eberhardt et al. (2021).

Three-dimensional receptor coordinates were retrieved from the Protein Data Bank (*Toxoplasma gondii* DHFR, PDB ID: 6AOG, chain B) wwPDB consortium (2019); Berman et al. (2003). To accurately simulate the native biological environment, the NADPH cofactor was retained within the catalytic pocket. Following the removal of crystallographic water and local minimization, polar hydrogens and Gasteiger charges were assigned to generate the required PDBQT files Bodun et al. (2025). The Vina search space was centered on the native CP6 ligand coordinates ($x = 3.689, y = 39.992, z = -62.818$) with grid box dimensions set to $20.0 \times 20.0 \times 20.0$ Å and an exhaustiveness parameter of 32. This configuration permitted unrestricted exploration of ligand orientations and flexible rotatable bonds, thereby preventing crystal-pose centering bias Trott and Olson (2010).

Prior to screening the machine learning candidates, the docking protocol underwent rigorous validation via re-docking of the co-crystallized CP6 ligand into the TgDHFR active site. The protocol successfully reproduced the native crystallographic binding mode, yielding a top-ranked pose with a centroid distance of 0.31 Å. This deviation falls well below the universally accepted 2.0 Å RMSD threshold, confirming the structural fidelity of the grid parameters and the reliability of the scoring function for this specific target.

Following validation, Vina evaluated the 3D conformers of the 10 top-ranked FDA candidates alongside the reference controls. Poses were ranked based on the predicted Gibbs free energy of binding (ΔG) in kcal/mol, where lower energies denote stronger thermodynamic stability Feng et al. (2023). To systematically evaluate the consensus between the ligand-based and structure-based approaches, we established strict quantitative thresholds. Compounds were classified as validated consensus hits only if they exhibited both high ML-predicted potency ($pIC_{50} > 6.5$) and strong docking affinity ($\Delta G < -8.0$ kcal/mol). Compounds meeting the ML threshold but failing the docking criteria were flagged as potential ML false positives. Finally, to statistically quantify the monotonic agreement between the two orthogonal methods without assuming linearity, a Spearman rank correlation analysis was conducted between the continuous pIC_{50} predictions and the Vina docking scores ($\rho = -0.394, p = 0.2600$).

2.7. In Silico Pharmacokinetic (ADMET) Profiling

Following consensus validation, a comprehensive predictive pharmacokinetic (ADMET) assessment evaluated the surviving candidates to reduce late-stage attrition risks. Candidates were first evaluated against Lipinski's Rule of Five criteria, strictly filtering out molecules exceeding 5 hydrogen bond donors, 10 hydrogen bond acceptors, or a calculated LogP of 5. To assess critical toxicity and absorption endpoints, custom machine learning models were trained using curated datasets from the Therapeutics Data Commons (TDC). Molecular structures were encoded as 2048-bit Morgan fingerprints (radius 2) and processed through Random Forest architectures comprising 100 estimators. A `RandomForestClassifier` trained on the TDC hERG dataset evaluated cardiac toxicity risk, advancing only compounds with a low probability of hERG channel blockade ($P < 0.5$). Concurrently, a `RandomForestRegressor` trained on the Caco2-Wang dataset predicted intestinal absorption, requiring candidates to demonstrate moderate-to-high permeability ($\log P_{app} > -5.15$). This final ADMET profiling ensured that the prioritized consensus hits represent not only highly efficient binders but also structurally validated, orally bioavailable starting points for therapeutic repurposing.

2.8. Computational Environment and Reproducibility

To guarantee algorithmic transparency and absolute computational reproducibility Rojas and Madanian (2026), all data curation, model training, and evaluation pipelines were programmatically implemented in Python (version 3.11.16). The computational environment relied on a standardized core software stack:

- **Cheminformatics and Featurization:** RDKit (version 2025.9.3) Feng et al. (2023) executed all molecular parsing, structural standardization, and descriptor generation (2D, 3D, and ECFP4).
- **Classical Machine Learning:** The nested cross-validation framework Krstajic et al. (2014); Calle et al. (2025), Bayesian hyperparameter optimization Baumann and Baumann (2014), and baseline algorithms leveraged `scikit-learn` (version 1.8.0) Vrba et al. (2026), `xgboost` (version 3.1.2) Wainer and Cawley (2021), `lightgbm` (version 4.6.0) Das et al. (2026), and `optuna` Meyer et al. (2025). Data manipulation and matrix operations were managed via `pandas` (version 2.3.3) and `numpy` (version 2.3.5) Eberhardt et al. (2021).
- **Deep Learning Infrastructure:** The D-MPNN graph models utilized `chemprop` (version 2.0) Graff et al. (2025); Heid et al. (2023) and `lightning`, with underlying tensor operations supported by PyTorch (`torch` $\geq 2.0.0$) Stegmüller et al. (2024).
- **Hardware Specifications:** All model training, hyperparameter tuning, and evaluation protocols ran on an Ubuntu 26.04 LTS workstation equipped with an Intel(R) Xeon(R) CPU E5-2695. The deliberate use of CPU-only acceleration explicitly benchmarks the computational accessibility and hardware demands of both classical and deep learning architectures in low-resource settings.

3. Results

3.1. Comparative Performance of Classical ML Across Feature Spaces

To establish a rigorous baseline for TgDHFR inhibitor prediction, we evaluated classical machine learning ensembles across four distinct molecular representations using a 25-fold nested cross-validation framework. The purely structural representation utilizing Morgan fingerprints (ECFP4, 2048 bits), devoid of additional physicochemical descriptors, yielded the highest predictive performance ($R^2 = 0.7417 \pm 0.0493$, MAE = 0.4841). Within this feature space, a stacked ensemble comprising LGBM+XGB+SVM emerged as the most frequently selected architecture during the inner cross-validation phase.

Escalating feature complexity failed to yield commensurate predictive gains. Integrating RDKit 2D descriptors with Morgan fingerprints (analogous to the original study’s “2D/FP” model) produced an R^2 of 0.7297 ± 0.0592 and an MAE of 0.4885. The most complex representation, incorporating 3D geometric descriptors derived from ETKDGv3 embedding and MMFF94s optimization (analogous to the “2D/3D/FP” model), attained an R^2 of 0.7348 ± 0.0614 and an MAE of 0.4835. A paired t-test comparing the purely structural Morgan fingerprint ensemble against the highly engineered 2D/3D/FP model revealed no statistically significant difference in explained variance ($t = 0.464$, $p = 0.6471$). Although the mean absolute error for the 2D/3D/FP descriptor set appeared marginally lower than that of the pure Morgan ensemble, a paired Wilcoxon signed-rank test across the 25 outer folds confirmed this variance lacks statistical significance ($W = 155.0$, $p = 0.8532$). These results indicate that conformational 3D featurization offers no measurable predictive advantage for this specific benchmark. Conversely, relying exclusively on 2D physicochemical descriptors severely degraded performance ($R^2 = 0.6785 \pm 0.0639$, MAE = 0.5301), confirming that topological fingerprints act as the primary drivers of predictive capacity.

To explicitly contextualize the computational cost-benefit ratio of advanced statistical techniques, we benchmarked our highest-performing classical model against the incremental progression reported in the original study (Table 1). Our classical stacking ensemble trained solely on Morgan features ($R^2 = 0.7417$) performed comparably to their unselected 2D/3D/FP baseline model ($R^2 = 0.75$). Because the original study reported point estimates without variance metrics, formal parametric comparisons assume the literature value as an absolute constant. Evaluated under a corrected resampled t-test, this variance lacks statistical significance ($t = -0.315$, $p = 0.7558$). Furthermore, our baseline 95% bootstrap confidence interval [0.7275, 0.8624] strictly encompasses the literature baseline, confirming statistical parity.

Table 1

Methodological comparison isolating the performance impact of architectural complexity, descriptor dimensionality, and data augmentation in TgDHFR bioactivity prediction.

Model Framework	Descriptors	Data Augmentation	Test R^2
Current Study (Classical Ensemble)	ECFP4 (2D)	None	0.7417
Baseline Karimova and Senanayake (2025)	2D/3D/FP	None	0.75
Augmented Final Model Karimova and Senanayake (2025)	2D/3D/FP (Selected)	Gaussian Noise	0.85

However, this baseline classical approach fell short of the original study’s final data-augmented deep neural network ensemble ($R^2 = 0.85$). The literature final model resides outside our baseline 95% CI, representing a statistically significant deviation ($t = -4.084$, $p = 0.0004$). These comparative results critically isolate the true source of predictive gains in this low-data regime. They demonstrate that achieving a competitive baseline performance ($R^2 \approx 0.75$) does not require computationally expensive 3D descriptors or complex neural network architectures; simple 2D fingerprints paired with classical machine learning algorithms suffice. Consequently, the performance gap between our baseline and the original study’s final model directly quantifies the impact of synthetic data augmentation rather than inherent architectural superiority.

3.2. Benchmarking Tree-Based Ensembles Against Graph Neural Networks

Graph Neural Networks (GNNs) dominate molecular property prediction by extracting representations directly from graph topologies. Yet, their predictive utility degrades sharply under data scarcity. To evaluate whether deep representation learning offers intrinsic advantages for the TgDHFR dataset (756 compounds), we benchmarked our optimal classical ensemble against a Directed Message Passing Neural Network (D-MPNN) using the Chemprop framework.

The D-MPNN yielded a cross-validated out-of-sample R^2 of 0.6818 ± 0.0555 and a Mean Absolute Error of 0.5329. Despite architectural sophistication, this network failed to match the classical stacking ensemble operating on pre-computed Morgan fingerprints ($R^2 = 0.7417 \pm 0.0493$). This performance deficit stems directly from data scarcity. Restricted to 756 training instances, the D-MPNN cannot learn graph convolutions capable of surpassing the predictive baseline established by fixed, topological ECFP4 descriptors.

This convergence plateau provides critical contextual validation for the methodologies in the original study Karimova and Senanayake (2025). The poor performance of the unaugmented D-MPNN explains the prior reliance on heavy statistical interventions—specifically, Gaussian data augmentation—to force predictive accuracy ($R^2 = 0.85$, $p < 0.001$). These empirical results substantiate our central thesis. Evaluated on unaugmented, data-limited chemical spaces, architectural complexity does not guarantee superior generalization. Classical tree-based ensembles utilizing fixed molecular descriptors remain the most robust, parsimonious, and computationally efficient alternative. Figure 2 details the comparative performance across all baseline models and the GNN.

3.3. Virtual Screening and Prioritization of FDA-Approved Candidates

To transition from theoretical benchmarking to practical drug discovery, we deployed the optimized classical ensemble (Morgan, $R^2 = 0.7417$) against a curated library of FDA-approved therapeutics. Before evaluating unknown chemical spaces, we quantified the predictive reliability of the pipeline using established TgDHFR inhibitors retained within the dataset. The algorithm successfully recovered these internal controls with minimal deviation. It assigned a predicted pIC_{50} of 6.4226 to pyrimethamine (experimental: 6.56, absolute error: 0.1374) and 5.6590 to trimethoprim (experimental: 5.57, absolute error: 0.0890). This tight empirical alignment confirms the ensemble’s capacity to interpolate bioactivity accurately within its defined applicability domain.

Following an unsupervised Isolation Forest filter to exclude structural outliers, the virtual screening protocol prioritized candidates based on absolute predicted potency and multi-objective ligand efficiency. Table 2 delineates the top ten prioritized compounds. Classic antifolate analogs—specifically folic acid, methotrexate, and pralatrexate—dominated the upper echelon of absolute affinity, all exceeding a predicted

Table 2

Top 10 FDA-approved candidates predicted by the classical ML model, ranked by predicted pIC_{50} and evaluated across standard ligand efficiency metrics.

Rank	Compound Name	Predicted pIC_{50}	LE	BEI	LLE	SEI
1	Folic Acid	8.1913	0.35	18.56	8.24	3.84
2	Methotrexate	7.7540	0.32	17.06	7.49	3.68
3	Pralatrexate	7.6354	0.30	15.99	6.65	3.68
4	Triamterene	6.8599	0.49	27.09	6.03	5.29
5	CID 336805	6.8052	0.23	12.08	7.35	2.69
6	Trimetrexate Glucuronate	6.8052	0.23	12.08	7.35	2.69
7	Bendamustine Hydrochloride	6.6463	0.38	16.84	2.96	11.39
8	Sumatriptan Succinate	6.4329	0.31	15.56	5.18	4.60
9	Pyrimethamine	6.4226	0.52	25.82	3.90	8.25
10	Chlorambucil	6.3704	0.46	20.94	2.99	15.71

pIC_{50} of 7.6. Absolute potency, however, frequently scales with molecular weight. This size dependency can artificially inflate binding scores and obscure the intrinsic thermodynamic quality of the underlying scaffold.

Correcting for physicochemical load via heavy-atom and molecular weight penalties revealed a distinct hierarchy of efficient binders. Smaller architectures demonstrated vastly superior atomic utilization. Pyrimethamine (LE = 0.52, BEI = 25.82) and the potassium-sparing diuretic triamterene (LE = 0.49, BEI = 27.09) emerged as the most structurally optimized candidates. Triamterene provides a particularly compelling repurposing profile. It achieves a robust predicted pIC_{50} of 6.8599 while maintaining exceptional ligand efficiency indices, indicating a highly complementary interaction with the TgDHFR catalytic pocket relative to its minimal steric footprint.

3.4. Integrated ML-docking consensus filtering and false-positive mitigation

Standalone virtual screening methodologies inherently generate false-positive artifacts. We implemented an orthogonal consensus filtering pipeline integrating ligand-based machine learning predictions with physics-based molecular docking to mitigate this structural vulnerability. Spearman’s rank correlation coefficient evaluated the global agreement between ML-predicted pIC_{50} values and AutoDock Vina binding energies across the prioritized subset ($N = 10$). This non-parametric approach ($\rho = -0.394$, $p = 0.2600$) robustly handles restricted sample sizes and isolates monotonic trends without assuming linear dependencies between topological bioactivity interpolation and thermodynamic scoring.

The negative correlation confirms that higher predicted potencies trend with stronger binding energies. The absence of statistical significance ($p > 0.05$) validates the methodological design. Two-dimensional ligand-based machine learning and three-dimensional physics-based docking capture fundamentally orthogonal aspects of molecular interaction. The ML ensemble interpolates bioactivity from topological patterns within the training distribution; Vina calculates thermodynamic stability via spatial complementarity and electrostatic interactions. Relying on either method in isolation exposes the screening process to distinct classes of false positives.

The consensus approach systematically identified and eliminated these methodological artifacts. Bisacodyl, for example, exhibited strong thermodynamic binding energies during structural validation ($\Delta G < -8.0$ kcal/mol). This represents a classic structure-based false positive driven by the spurious accommodation of flexible, hydrophobic scaffolds within the binding pocket. The classical ML model correctly assigned a low predicted affinity to this compound, vetoing its progression. Dual validation prevents the misallocation of experimental resources toward single-method screening artifacts.

Orthogonal docking isolated 4 consensus hits from the 10 evaluated compounds satisfying both the ML potency threshold and stringent docking energy criteria (including the novel candidates Folic Acid and Triamterene, alongside the control Pyrimethamine). The primary consensus candidate, Pralatrexate, demonstrated a strong ML-predicted pIC_{50} of 7.6354 alongside a favorable docking score of -9.43 kcal/mol.

This lightweight, classical pipeline matches the statistical performance of unaugmented deep learning models while discovering biologically viable, physics-validated inhibitors. Isolating high-confidence repurposing candidates without synthetic data generation or 3D feature engineering yields a superior computational cost-benefit ratio for practical drug discovery in low-data regimes.

4. Discussion

4.1. The Role of Explicit Inductive Bias in Small-Scale Molecular Datasets

Graph Neural Networks theoretically offer superior representational capacity by learning molecular features directly from graph topology, thereby bypassing the need for manual feature engineering. However, this architectural flexibility demands extensive training data to converge on generalizable chemical representations. In the context of the TgDHFR dataset, which comprises only 756 compounds, the GNN architecture failed to leverage its theoretical superiority, plateauing at a predictive performance statistically indistinguishable from classical machine learning approaches.

This convergence failure highlights the critical role of explicit inductive bias in low-data regimes. The fundamental lack of explicit chemical priors in deep representation learning forces these architectures to construct a continuous latent space entirely from scratch. With fewer than 1,000 training instances, the resulting latent space density is severely restricted, creating a sparse manifold where interpolation between data points becomes highly unstable. To counteract this sparsity and force convergence, researchers are frequently compelled to inject artificial variance into the training pipeline. As observed in the original study Karimova and Senanayake (2025), deep neural networks on this specific dataset required extensive Gaussian noise-based data augmentation to artificially inflate latent space density and achieve high predictive accuracy.

Classical tree-based ensembles operating on pre-computed Morgan fingerprints bypass the need for end-to-end representation learning by utilizing fixed, domain-specific descriptors. These fingerprints encode established chemical heuristics—specifically, circular topological substructures—directly into the feature space. For small-scale datasets, this explicit pre-computation acts as a natural regularizer, rigidly constraining the hypothesis space and preventing the model from memorizing noise. Our results explicitly demonstrate that exploiting the inductive bias inherent to traditional structural fingerprints entirely eliminates the necessity for computationally burdensome data augmentation. When data volume is insufficient to natively train deep graph convolutions and populate a dense latent space, deterministic feature engineering remains a vastly more reliable, parsimonious, and elegant strategy for molecular property prediction.

4.2. Dimensionality, Conformational Overhead, and Model Efficiency

Beyond predictive accuracy, the computational efficiency of molecular representations dictates their practical utility in high-throughput virtual screening. Incorporating 3D geometric descriptors necessitates extensive conformational sampling and energy minimization (e.g., via ETKDGV3 embedding and MMFF94s optimization). This requirement introduces a substantial computational bottleneck, termed conformational overhead, which scales prohibitively as screening libraries expand into the millions of compounds.

Our empirical evaluation revealed that this added computational cost failed to yield commensurate predictive gains. The highly engineered 2D/3D/FP representation achieved an R^2 of 0.7348, a performance statistically indistinguishable from the purely structural Morgan fingerprint model ($R^2 = 0.7417$, $p = 0.6471$). This parity indicates that the relevant pharmacophoric features governing TgDHFR inhibition are adequately captured by 2D topological connectivity alone.

We frame this result as the advantage of parsimony. By deliberately omitting complex 3D descriptors and synthetic data augmentation, the optimized classical pipeline circumvents severe computational expenditures during feature preprocessing and model training. Deploying computationally expensive 3D descriptors for this specific target introduces unnecessary dimensionality and processing latency without any tangible improvement in model fidelity. These findings reinforce the utility of lightweight, 2D fingerprint-based ensembles as an efficient, highly scalable paradigm for virtual screening in low-data environments.

4.3. Methodological Scope, Limitations, and Translational Potential

The predictive reliability of any quantitative structure-activity relationship (QSAR) model is inherently bounded by the chemical diversity of its training corpus. Given the dataset size of 756 compounds,

the applicability domain of our optimized ensemble is strictly restricted to the structural scaffolds and physicochemical properties represented within the curated TgDHFR data. Consequently, extrapolating predictions to highly divergent chemotypes or novel synthetic spaces carries an increased risk of reduced predictive confidence.

A critical vulnerability intrinsic to 2D circular fingerprints in low-data regimes is their susceptibility to "activity cliffs"—pairs of structurally similar molecules that exhibit radically divergent bioactivities due to subtle chemical modifications. For example, a single bioisosteric substitution or methylation can alter a critical hydrogen bond or steric fit within the binding pocket. Because Morgan fingerprints map topological neighborhoods into bit vectors without explicitly encoding 3D spatial conformation or electrostatic field potentials, they risk mapping active and inactive cliff-forming analogs to virtually identical representations. In this study, we strategically mitigated this limitation through our downstream multi-tier architecture. The orthogonal integration of physics-based molecular docking explicitly evaluates 3D spatial complementarity and steric clashes, acting as a structural filter that detects activity cliffs otherwise invisible to pure 2D topological models.

An additional methodological constraint involves the extensive architecture search executed within the inner optimization loop. Evaluating up to 41 distinct stacking combinations across six regressor families introduces a latent risk of model selection bias. While the nested cross-validation framework mathematically isolates this optimization phase from the outer test folds—thereby mitigating spurious performance inflation—it does not entirely eliminate architectural instability. To transparently quantify this variance, Supplementary Table S1 details the selection frequency of each meta-ensemble across the 25 outer folds. Within the optimal Morgan fingerprint space, the LGBM+XGB+SVM configuration emerged as the most stable architecture, yet it was selected in only 12.0% of the iterations. This distribution confirms that while the nested design successfully prevents test-set leakage, the inherent data sparsity of the 756-compound benchmark inevitably restricts the convergence of the inner loop onto a single, universally dominant stacking hierarchy.

Despite these boundary conditions, the programmatic pipeline developed in this study offers substantial translational utility. By establishing a computationally lightweight framework, this approach functions as a highly efficient triage mechanism, rapidly filtering extensive libraries of approved drugs to isolate a tractable subset of candidates. This targeted prioritization minimizes the computational overhead required for subsequent physics-based simulations and systematically reduces the attrition rate in downstream *in vitro* and *in vivo* experimental validations.

4.4. Conclusions and Future Directions

This study re-evaluates the methodological prerequisites for molecular property prediction in low-data environments ($N = 756$), yielding three principal findings. First, classical machine learning ensembles operating on 2D topological fingerprints provide sufficient explicit inductive bias ($R^2 = 0.7417 \pm 0.0493$) to surpass the predictive utility of unaugmented deep learning architectures ($R^2 = 0.6818$) and computationally expensive 3D descriptors. Second, rigorous nested cross-validation remains imperative to eliminate optimistic performance bias and accurately quantify out-of-sample generalization. Finally, integrating ligand-based statistical predictions with physics-based molecular docking establishes a robust consensus filter. This orthogonal validation mitigates the false-positive rates inherent to isolated screening methodologies, isolating 4 structurally viable candidates—most notably Pralatrexate—for therapeutic repurposing.

By executing established benchmarks under leak-free validation protocols, we demonstrate that competitive predictive accuracy is attainable without the severe computational overhead of synthetic data augmentation or 3D conformational modeling. These empirical results consolidate the principle of parsimony as a foundational heuristic for computational drug discovery. The resulting framework provides an accessible, reproducible, and computationally efficient pipeline for rapid therapeutic repurposing in data-limited regimes.

Data and Code Availability

All source code, Python scripts for dataset processing, nested cross-validation, model training, and docking validation protocols, as well as the benchmark datasets, are publicly available in the GitHub repository at the following [link](#)

Declaration of Generative AI's use

During the preparation of this work, the author(s) utilized generative artificial intelligence technologies (Large Language Models) solely to assist with language polishing, structural formatting, and improving the overall readability of the manuscript. After using these tools, the author(s) thoroughly reviewed, edited, and validated all generated text. The author(s) take full and sole responsibility for the final content, scientific integrity, and accuracy of this publication. No AI technologies were used to generate data, perform statistical analyses, or draw scientific conclusions.

Acknowledgments

This work was supported in part by ANID FONDECYT Regular No. 1241977, FONDECYT Iniciación No. 11261195, and ANID CPS-RTC CIA250016 in Chile.

CRedit authorship contribution statement

Andrey Vinajera-Zamora: Conceptualization, Methodology, Formal analysis, Software, Validation, Investigation, Writing – original draft, Visualization. **Samuel Montejo-Sánchez:** Validation, Writing – review & editing, Supervision. **Raul Arias-Carrasco:** Validation, Supervision, Writing – review & editing, Visualization. **Juan Castillo-Garit:** Validation, Writing – review & editing.

References

- Abad-Zapatero, C., Metz, J.T., 2005. Ligand efficiency indices as guideposts for drug discovery. *Drug Discovery Today* 10, 464–469. doi:10.1016/S1359-6446(05)03386-6.
- Azad, A.K., 2026. Chapter 2 - ai strategies for drug discovery and bioactivity prediction: opportunities and challenges, in: Chen, J.T. (Ed.), *Revolutionizing Drug Development*. Academic Press, pp. 13–24. doi:10.1016/B978-0-443-34059-8.00013-0.
- Bansal, A., Sharma, A.R., Chakraborty, A., Sunkaria, A., Jain, A., 2025. Machine learning-based bioactivity prediction and descriptor-guided rational design of amyloid-b aggregation inhibitors. *ACS Chemical Neuroscience* doi:10.1021/acscchemneuro.5c00649.
- Basit Iqbal, A., Assad, A., Bhat, B., Macha, M.A., Shah, S.Z.A., 2025. Mpnn-cwexplainer: An enhanced deep learning framework for hiv drug bioactivity prediction with class-weighted loss and explainability. *Life Sciences* 378, 123835. doi:10.1016/j.lfs.2025.123835.
- Baumann, D., Baumann, K., 2014. Reliable estimation of prediction errors for qsar models under model uncertainty using double cross-validation. *Journal of Cheminformatics* 6. doi:10.1186/s13321-014-0047-1.
- Berman, H., Henrick, K., Nakamura, H., 2003. Announcing the worldwide protein data bank. *Nature structural & molecular biology* 10, 980–980.
- Bodun, D.S., Ibrahim, I.O., Bashiru, M., Abolade, R.O., Ndarawit, W.K., John-Joy Owolade, A., Temitope, A., Somtochukwu, I.J., Olugbogi, E.A., Samadhi, K.G.A., Oduselu, G., Sharif, S., Omoboyowa, D.A., Ajani, O.O., 2025. De novo design and bioactivity prediction of mitotic kinesin eg5 inhibitors using mpnn and lstm-based transfer learning. *Computers in Biology and Medicine* 195, 110672. doi:10.1016/j.combiomed.2025.110672.
- Calle, P., Bates, A., Reynolds, J.C., Liu, Y., Cui, H., Ly, S., Wang, C., Zhang, Q., de Armendi, A.J., Shettar, S.S., Fu, K.M., Tang, Q., Pan, C., 2025. Integration of nested cross-validation, automated hyperparameter optimization, high-performance computing to reduce and quantify the variance of test performance estimation of deep learning models. *Computer Methods and Programs in Biomedicine* 272, 109063. doi:10.1016/j.cmpb.2025.109063.
- Cawley, G.C., Talbot, N.L., 2010. On over-fitting in model selection and subsequent selection bias in performance evaluation. *The Journal of Machine Learning Research* 11, 2079–2107.
- Cherkasov, A., Muratov, E.N., Fourches, D., Varnek, A., Baskin, I.I., Cronin, M., Dearden, J., Gramatica, P., Martin, Y.C., Todeschini, R., Consonni, V., Kuz'min, V.E., Cramer, R., Benigni, R., Yang, C., Rathman, J., Terfloth, L., Gasteiger, J., Richard, A., Tropsha, A., 2013. Qsar modeling: Where have you been? where are you going to? *Journal of Medicinal Chemistry* 57, 4977–5010. doi:10.1021/jm4004285.
- wwPDB consortium, 2019. Protein data bank: the single global archive for 3d macromolecular structure data. *Nucleic Acids Research* 47, D520–D528. doi:10.1093/nar/gky949.

- Correa, J., Tavakoli, H., Vogel, S., Gebbers, R., 2025. Overfitting due to data leakage in soil sensor calibration: Examples from lab-based and in-situ soil nir spectroscopy. *Computers and Electronics in Agriculture* 239, 110920. doi:10.1016/j.compag.2025.110920.
- Dalmau, D., Sigman, M.S., Alegre-Requena, J.V., 2025. Machine learning workflows beyond linear models in low-data regimes. *Chemical Science* 16, 8555–8560. doi:10.1039/d5sc00996k.
- Dar, S.U.H., Şaban Öztürk, Özbey, M., Oguz, K.K., Çukur, T., 2023. Parallel-stream fusion of scan-specific and scan-general priors for learning deep mri reconstruction in low-data regimes. *Computers in Biology and Medicine* 167, 107610. doi:10.1016/j.combiomed.2023.107610.
- Das, R., Sinnarasan, V.S.P., Paul, D., Sheikh, M.M.R., Venkatesan, A., 2026. Covipredx: Ai-driven bioactivity prediction for small molecule therapeutics against sars-cov-2. *Computational Biology and Chemistry* 124, 109167. doi:10.1016/j.combiolchem.2026.109167.
- Eberhardt, J., Santos-Martins, D., Tillack, A.F., Forli, S., 2021. Autodock vina 1.2. 0: new docking methods, expanded force field, and python bindings. *Journal of chemical information and modeling* 61, 3891.
- Espley, S.G., Farrar, E.H.E., Buttar, D., Tomasi, S., Grayson, M.N., 2023. Machine learning reaction barriers in low data regimes: a horizontal and diagonal transfer learning approach††electronic supplementary information (esi) available: Additional details on generation of sqm and dft data, ml features, ml/tl analysis and model performance evaluation. see doi: 10.1039/d3dd00085k. *Digital Discovery* 2, 941–951. doi:10.1039/d3dd00085k.
- Feng, H., Jiang, J., Wei, G.W., 2023. Machine-learning repurposing of drugbank compounds for opioid use disorder. *Computers in Biology and Medicine* 160, 106921. doi:10.1016/j.combiomed.2023.106921.
- Gao, K., Nguyen, D.D., Sresht, V., Mathiowetz, A.M., Tu, M., Wei, G.W., 2020. Are 2d fingerprints still valuable for drug discovery? *Physical Chemistry Chemical Physics* 22, 8373–8390. doi:10.1039/d0cp00305k.
- Gerz, F., Jelali, M., 2026. Towards reliable data augmentation in machine learning: Practices to prevent data leakage. *Smart Agricultural Technology* 15, 102462. doi:10.1016/j.atech.2026.102462.
- Goodarzi, M., Heyden, Y.V., Funar-Timofei, S., 2013. Towards better understanding of feature-selection or reduction techniques for quantitative structure–activity relationship models. *TrAC Trends in Analytical Chemistry* 42, 49–63. doi:10.1016/j.trac.2012.09.008.
- Graff, D.E., Morgan, N.K., Burns, J.W., Doner, A.C., Li, B., Li, S.C., Manu, J., Menon, A., Pang, H.W., Wu, H., Zalte, A.S., Zheng, J.W., Coley, C.W., Green, W.H., Greenman, K.P., 2025. Chemprop v2: An efficient, modular machine learning package for chemical property prediction. *Journal of Chemical Information and Modeling* 66, 28–33. doi:10.1021/acs.jcim.5c02332.
- Hawkins, D.M., 2004. The problem of overfitting. *Journal of chemical information and computer sciences* 44, 1–12.
- Heid, E., Greenman, K.P., Chung, Y., Li, S.C., Graff, D.E., Vermeire, F.H., Wu, H., Green, W.H., McGill, C.J., 2023. Chemprop: A machine learning package for chemical property prediction. *Journal of Chemical Information and Modeling* 64, 9–17. doi:10.1021/acs.jcim.3c01250.
- Hopkins, A.L., Groom, C.R., Alex, A., 2004. Ligand efficiency: a useful metric for lead selection. *Drug Discovery Today* 9, 430–431. doi:10.1016/S1359-6446(04)03069-7.
- Hopkins, A.L., Keserü, G.M., Leeson, P.D., Rees, D.C., Reynolds, C.H., 2014. The role of ligand efficiency metrics in drug discovery. *Nature reviews Drug discovery* 13, 105–121.
- Iwuala, B.N., Aliyu, A.B., Ayo, R.G.O., Hamza, A.N., Madumelu, M., Habila, J.D., 2025. Quantitative structure-activity relationship (qsar) studies of dihydroorotate dehydrogenase inhibitors of plasmodium falciparum for the combat of malaria resistance using machine learning models classification approach. In *Silico Research in Biomedicine* 1, 100019. doi:10.1016/j.insr.2025.100019.
- Iwuala, B.N., Madumelu, M., Aliyu, A.B., Ayo, R.G.O., Hamza, A.N., Habila, J.D., 2026. Optimizing antimalarial discovery: A robust machine learning framework for chalcone bioactivity prediction using ensemble classifiers and oversampling strategies. *Scientific African* 32, e03393. doi:10.1016/j.sciaf.2026.e03393.
- John, K., Saurette, D.D., Heung, B., 2025. The problematic case of data leakage: A case for leave-profile-out cross-validation in 3-dimensional digital soil mapping. *Geoderma* 455, 117223. doi:10.1016/j.geoderma.2025.117223.
- Karimova, N.V., Senanayake, R.D., 2025. Ai-driven prediction of drug activity against toxoplasma gondii: Data augmentation and deep neural networks for limited datasets. *Artificial Intelligence Chemistry* 3, 100084. doi:10.1016/j.aichem.2025.100084.
- Király, P., Kiss, R., Kovács, D., Ballaj, A., Tóth, G., 2022. The relevance of goodness-of-fit, robustness and prediction validation categories of oecd-qsar principles with respect to sample size and model type. *Molecular Informatics* 41. doi:10.1002/minf.202200072.
- Konstantakos, S., Cani, J., Mademlis, I., Chalkiadaki, D.I., Asano, Y.M., Gavves, E., Papadopoulos, G.T., 2025. Self-supervised visual learning in the low-data regime: A comparative evaluation. *Neurocomputing* 620, 129199. doi:10.1016/j.neucom.2024.129199.
- Krstajic, D., Buturovic, L.J., Leahy, D.E., Thomas, S., 2014. Cross-validation pitfalls when selecting and assessing regression and classification models. *Journal of cheminformatics* 6, 10.
- Landrum, G., Tosco, P., Kelley, B., Rodriguez, R., Cosgrove, D., Vianello, R., Gedeck, P., Jones, G., Kawashima, E., Nealschneider, D., et al., 2026. rdkit/rdkit: 2025_09_6 (q3 2025) release. Zenodo .
- Lewis, M.J., Spiliopoulou, A., Goldmann, K., Pitzalis, C., McKeigue, P., Barnes, M.R., 2023. nestedcv: an r package for fast implementation of nested cross-validation with embedded feature selection designed for transcriptomics and high-dimensional data. *Bioinformatics Advances* 3, vbad048. doi:10.1093/bioadv/vbad048.

- Liu, Y., Liu, R., Duan, J., Wang, L., Zhang, X., Li, G., 2023. Deep generative fuel design in low data regimes via multi-objective imitation. *Chemical Engineering Science* 274, 118686. doi:10.1016/j.ces.2023.118686.
- Matsuzaka, Y., Uesawa, Y., 2023. Ensemble learning, deep learning-based and molecular descriptor-based quantitative structure–activity relationships. *Molecules* 28. doi:10.3390/molecules28052410.
- Meyer, T.N., Andreeva, O., Weiss, R.D., Ding, W., Shen, I., Wang, C., Chen, P., Dagnew, T.M., 2025. Enhancing neuromolecular imaging classification in low-data regimes with generative machine learning: A case study in hdac pet/mr imaging of alcohol use disorder. *Neuroscience Informatics* 5, 100225. doi:10.1016/j.neuri.2025.100225.
- Nadeau, C., Bengio, Y., 1999. Inference for the generalization error, in: Solla, S., Leen, T., Müller, K. (Eds.), *Advances in Neural Information Processing Systems*, MIT Press. URL: https://proceedings.neurips.cc/paper_files/paper/1999/file/7d12b66d3df6af8d429c1a357d8b9e1a-Paper.pdf.
- Oulladji, L., Saadallah, M., Guellil, Z., Abbad, H., Bekhti, H., 2025. Ai-driven molecule generation and bioactivity prediction: A multi-model approach combining vae, graph and language-based neural networks. *Computational Biology and Chemistry* 119, 108532. doi:10.1016/j.compbiolchem.2025.108532.
- Parvande, S., Yeh, H.W., Paulus, M.P., McKinney, B.A., 2020. Consensus features nested cross-validation. *Bioinformatics* 36, 3093–3098. doi:10.1093/bioinformatics/btaa046.
- Riniker, S., Landrum, G.A., 2015. Better informed distance geometry: Using what we know to improve conformation generation. *Journal of Chemical Information and Modeling* 55, 2562–2574. doi:10.1021/acs.jcim.5b00654.
- Rogers, D., Hahn, M., 2010. Extended-connectivity fingerprints. *Journal of Chemical Information and Modeling* 50, 742–754. doi:10.1021/ci100050t.
- Rojas, F., Madanian, S., 2026. Bias in developing ai medical diagnosis: Impact of sequential ct scan slices on data leakage in lung cancer. *Informatics in Medicine Unlocked* 65, 101799. doi:10.1016/j.imu.2026.101799.
- Rücker, C., Rücker, G., Meringer, M., 2007. y-randomization and its variants in qspr/qsar. *Journal of Chemical Information and Modeling* 47, 2345–2357. doi:10.1021/ci700157b.
- Shalit Peleg, H., Milo, A., 2023. Small data can play a big role in chemical discovery. *Angewandte Chemie* 135, e202219070.
- Shultz, M.D., 2013. Setting expectations in molecular optimizations: Strengths and limitations of commonly used composite parameters. *Bioorganic & Medicinal Chemistry Letters* 23, 5980–5991. doi:10.1016/j.bmcl.2013.08.029.
- Stegmüller, T., Abbet, C., Bozorgtabar, B., Clarke, H., Petignat, P., Vassilakos, P., Thiran, J.P., 2024. Self-supervised learning-based cervical cytology for the triage of hpv-positive women in resource-limited settings and low-data regime. *Computers in Biology and Medicine* 169, 107809. doi:10.1016/j.compbiomed.2023.107809.
- Tiwari, S., Madra-Gackowska, K., Gackowski, M., Szeleszczuk, L., 2026. Evaluating the incremental value of three-dimensional conformer descriptors for blood–brain barrier permeability prediction using the b3db benchmark dataset. *Pharmaceutics* 18. doi:10.3390/pharmaceutics18081018.
- Trott, O., Olson, A.J., 2010. Autodock vina: Improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading. *Journal of Computational Chemistry* 31, 455–461. doi:10.1002/jcc.21334.
- Varma, S., Simon, R., 2006. Bias in error estimation when using cross-validation for model selection. *BMC bioinformatics* 7, 91.
- Veetil, I.K., Chowdary, D.E., Chowdary, P.N., Sowmya, V., Gopalakrishnan, E., 2024. An analysis of data leakage and generalizability in mri based classification of parkinson's disease using explainable 2d convolutional neural networks. *Digital Signal Processing* 147, 104407. doi:10.1016/j.dsp.2024.104407.
- Vrba, J., Steinbach, J., Jirsa, T., Homma, N., 2026. Feature scaling induced data leakage quantification in machine learning-based voice pathology detection. *Applied Soft Computing* 199, 115349. doi:10.1016/j.asoc.2026.115349.
- Wainer, J., Cawley, G., 2021. Nested cross-validation when selecting classifiers is overzealous for most practical applications. *Expert Systems with Applications* 182, 115222. doi:10.1016/j.eswa.2021.115222.
- Winkler, D.A., 2026. Are we underestimating overfitting? *Journal of Chemical Information and Modeling* 66, 6218–6224. doi:10.1021/acs.jcim.6c00518.
- Xie, W., Wiriyanattanakul, S., Rungrotmongkol, T., Shi, L., Wiriyanattanakul, A., Maitarad, P., 2023. Rational design of a low-data regime of pyrrole antioxidants for radical scavenging activities using quantum chemical descriptors and qsar with the ga-mlr and ann concepts. *MOLECULES* 28. doi:10.3390/molecules28041596.
- Xiong, Z., Cui, Y., Liu, Z., Zhao, Y., Hu, M., Hu, J., 2020. Evaluating explorative prediction power of machine learning algorithms for materials discovery using k-fold forward cross-validation. *Computational Materials Science* 171, 109203. doi:10.1016/j.commatsci.2019.109203.
- Yang, K., Swanson, K., Jin, W., Coley, C., Eiden, P., Gao, H., Guzman-Perez, A., Hopper, T., Kelley, B., Mathea, M., et al., 2019. Analyzing learned molecular representations for property prediction. *Journal of chemical information and modeling* 59, 3370.
- Zhang, Y., Bai, X., 2024. Geometry-augmented molecular representation learning for property prediction. *IEEE/ACM Transactions on Computational Biology and Bioinformatics* 21, 1518–1528. doi:10.1109/TCBB.2024.3402337.
- Özçelik, R., Grisoni, F., 2024. A hitchhiker's guide to deep chemical language processing for bioactivity prediction††electronic supplementary information (esi) available. see doi: 10.1039/d4dd00311j. *Digital Discovery* 4, 316–325. doi:10.1039/d4dd00311j.

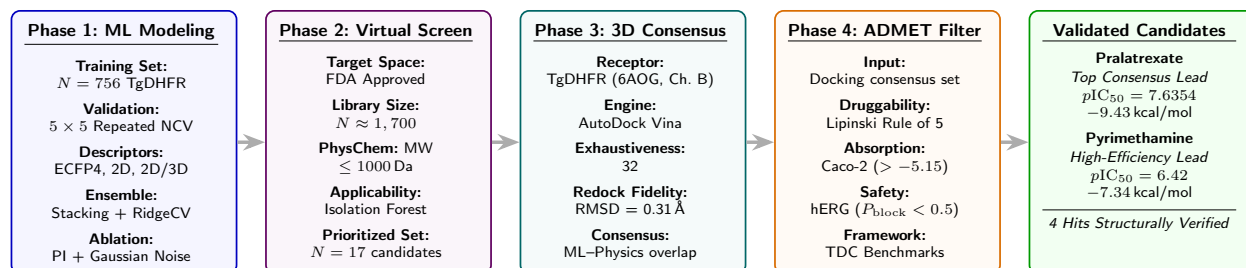


Figure 1: Schematic overview of the multi-tier computational repurposing and validation pipeline. The hierarchical funnel integrates: (1) 2D ligand-based representation benchmarking under nested cross-validation, (2) applicability domain-constrained screening of FDA-approved compounds, (3) orthogonal physics-based consensus validation via molecular docking (AutoDock Vina), and (4) in silico ADMET profiling (Caco-2 permeability and hERG cardiotoxicity) to isolate clinically viable TgDHFR candidates.

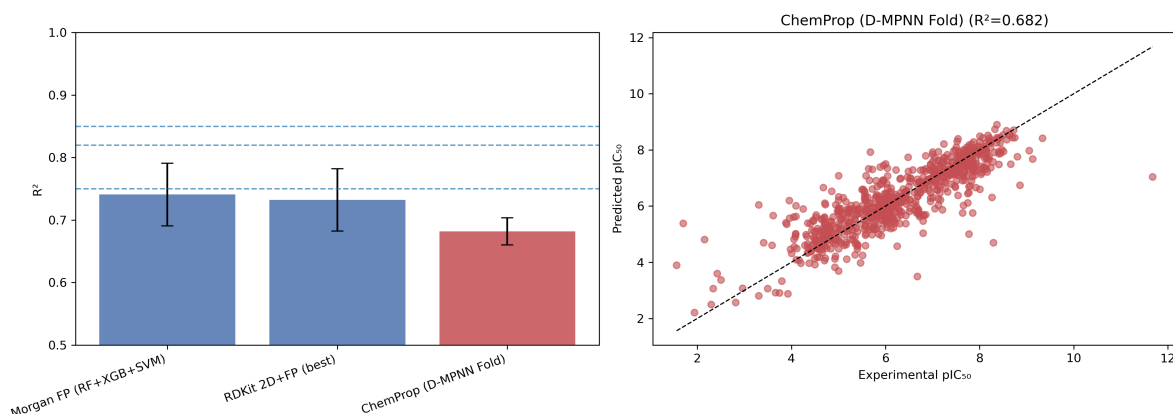


Figure 2: Comparative predictive performance of classical machine learning ensembles, the Chemprop D-MPNN, and the original study's baseline and final models. The scatter plot details the predicted versus experimental pIC_{50} values for the D-MPNN.